

Canadian International College

Business School

DEPARTMENT OF Business Technology



PROJECT REPORT

AUTOPENT AI

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Abdulrahman Gamal – Mohamed Hussien – Nada Mohamed – Rania Omar**

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Supervisor: Dr. Ahmed Gaber Abuabdallah

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Acknowledgment

First and foremost, we would like to express our deepest gratitude and sincere appreciation to our supervisor, **Dr. Ahmed Gaber Abuabdullah**, for his continuous support, invaluable guidance, encouragement, and constructive feedback throughout the development of this graduation project. His expertise, patience, and dedication played a significant role in helping us overcome challenges and achieve the objectives of this work. We are truly grateful for the time and effort he devoted to reviewing our progress, providing insightful recommendations, and inspiring us to maintain high academic and professional standards. His guidance not only contributed to the successful completion of this project but also enriched our knowledge and understanding of the field. We would also like to extend our thanks to the faculty members of our department for providing us with the educational foundation and resources necessary for carrying out this project. Their support and dedication have been instrumental in our academic journey. Finally, we would like to express our heartfelt appreciation to our families and friends for their continuous encouragement, patience, and unwavering support throughout our studies and during the completion of this project. Their motivation and belief in our abilities have been a constant source of strength.

ABSTRACT

With the rapid increase in cyber threats and the evolving digital landscape, ensuring web application security has become a critical challenge for organizations and developers. Traditional manual penetration testing is often time-consuming, resource-intensive, and requires specialized expertise, while existing automated tools typically suffer from fragmentation and complexity. These tools often generate passive technical reports that highlight risks without offering clear, actionable solutions, leaving systems vulnerable to exploitation for longer periods.

This project presents "AutoPent AI," an automated Vulnerability Scanner and Remediation Tool designed to bridge the gap between vulnerability detection and code correction. The system features a responsive, user-friendly interface developed using React.js, coupled with a robust backend controller built in Python to orchestrate complex security operations. The core engine seamlessly integrates powerful open-source security standards, including OWASP ZAP for web application scanning, Nmap for network port analysis, and OpenVAS for infrastructure vulnerability management, into a single efficient engine.

A key innovation of this system is its AI-driven Remediation Module. Unlike traditional scanners, this system utilizes artificial intelligence to analyze vulnerability findings and generate specific, syntax-correct code fixes and patching suggestions. Developed following the Agile SDLC methodology to allow for iterative improvements and continuous feedback, this approach transforms the penetration testing process from a passive reporting task into an active, intelligent defense mechanism, enabling developers to secure their applications efficiently and effectively.

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1. Project Title

1.1. Problem Statement

Web applications and network systems face increasing cybersecurity threats, while manual penetration testing remains costly and time-consuming. This project addresses the need for an automated solution that not only detects security vulnerabilities but also provides clear remediation guidance, helping organizations reduce risks and strengthen their security posture more efficiently.

1.2. Objectives

The main objectives of this project are:

1. To develop an automated web-based vulnerability scanner using **PHP, JavaScript, and Python**.
2. To integrate industry-standard security tools (Burp Suite, Nmap, OpenVAS) into a single efficient engine.
3. To bridge the gap between security and development by providing **AI-generated code fixes** for detected vulnerabilities.
4. To implement a secure user environment with authentication (Login/Signup) and a history log to track previous scans.
5. To produce a working prototype that demonstrates local scanning capabilities, with plans to expand to advanced AI chat features in the second phase.
6. To ensure ethical and legal usage by limiting scans to authorized targets only.

2. ABOUT THE PROJECT

2.1. Methodology and Tools

The project follows the **Agile SDLC methodology**, which supports iterative development and flexibility. This approach allows the team to adapt to new requirements and technical challenges that may arise during the implementation of complex security tools. The development process is divided into managed sprints, ensuring continuous testing and feedback integration. This is particularly suitable for the project's two-term structure:

- **Phase 1 (Current Term):** Focuses on tool research, prototype development, core scanning engine implementation, and basic frontend-backend integration.
- **Phase 2 (Next Term):** Focuses on integrating the advanced AI Remediation Module, implementing the chat feature, and finalizing the system for deployment.

2.2. Budgeting

2.3. Timeline and Duration

2.4. Conclusion

This project developed a web-based Penetration Testing System that automates vulnerability assessment for websites and web applications. By integrating multiple security tools and technologies, the system efficiently detects vulnerabilities, generates security reports, and provides remediation recommendations. The results demonstrated its effectiveness in enhancing web application security and supporting organizations in protecting their digital assets.

2.5. Figures and Tables

Figures:

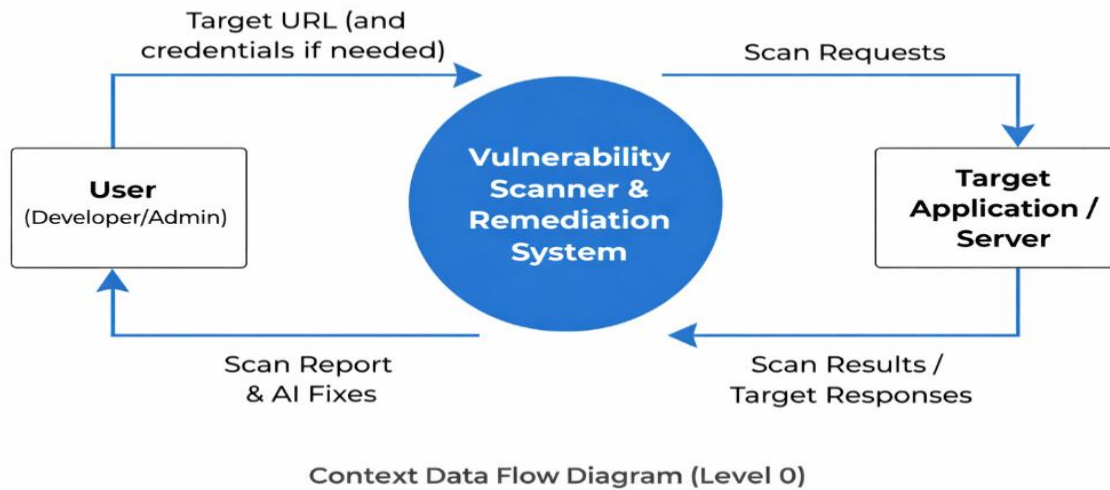


Figure 3.1: Context Data Flow Diagram (DFD)

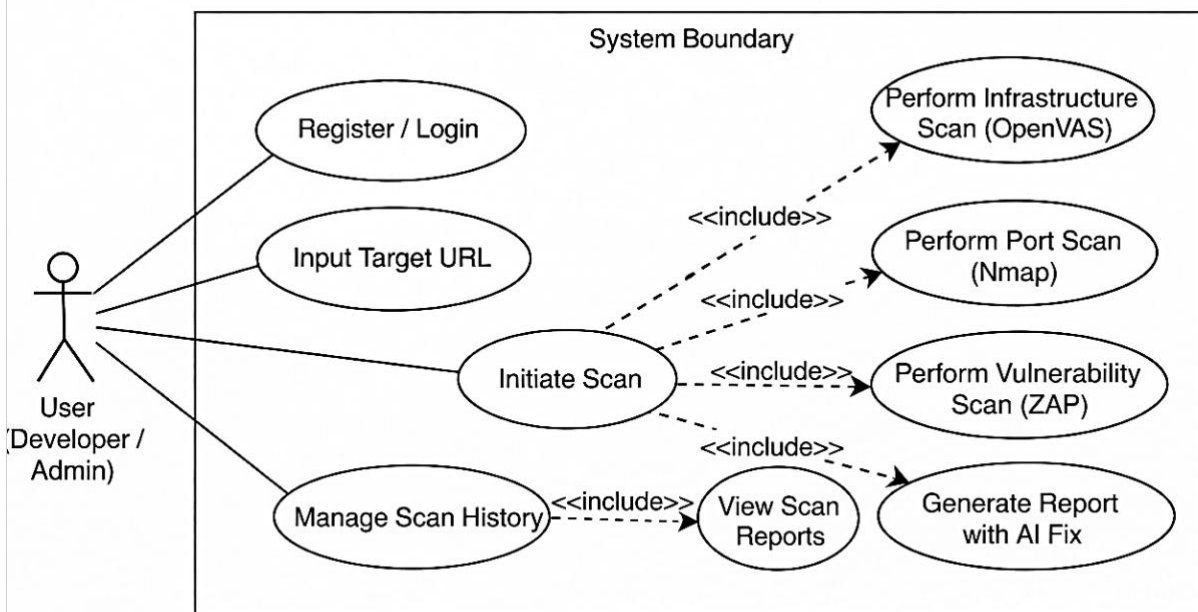


Figure 3.2: Use Case Diagram

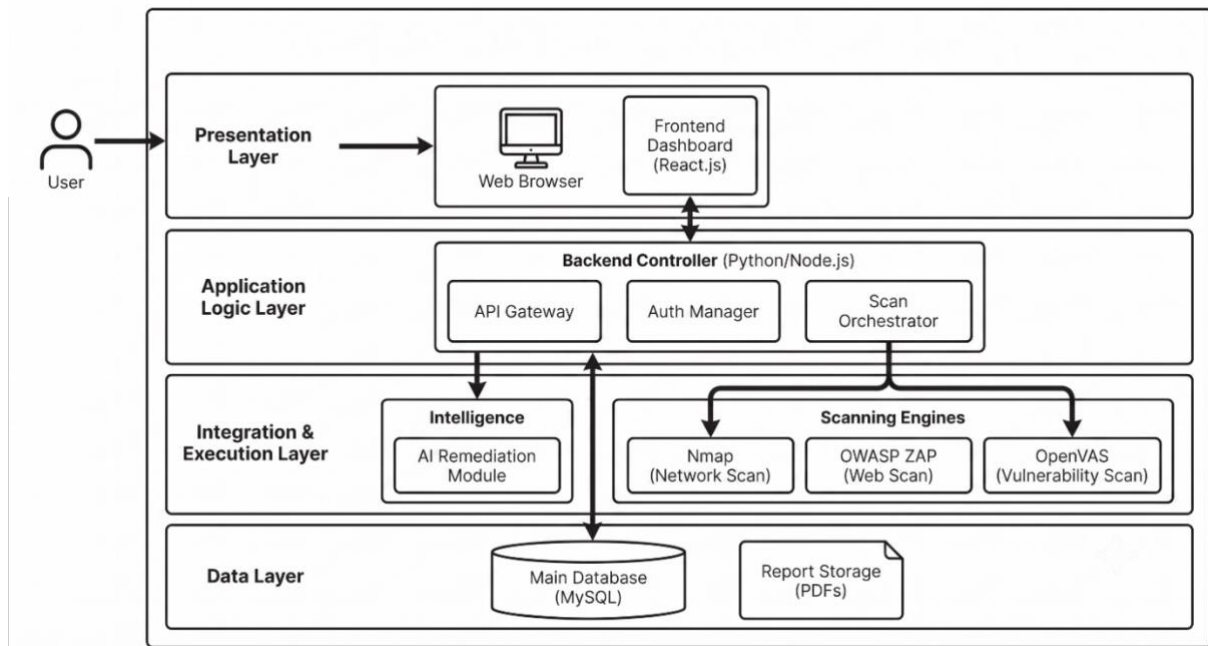


Figure 4.1: System Architecture Diagram

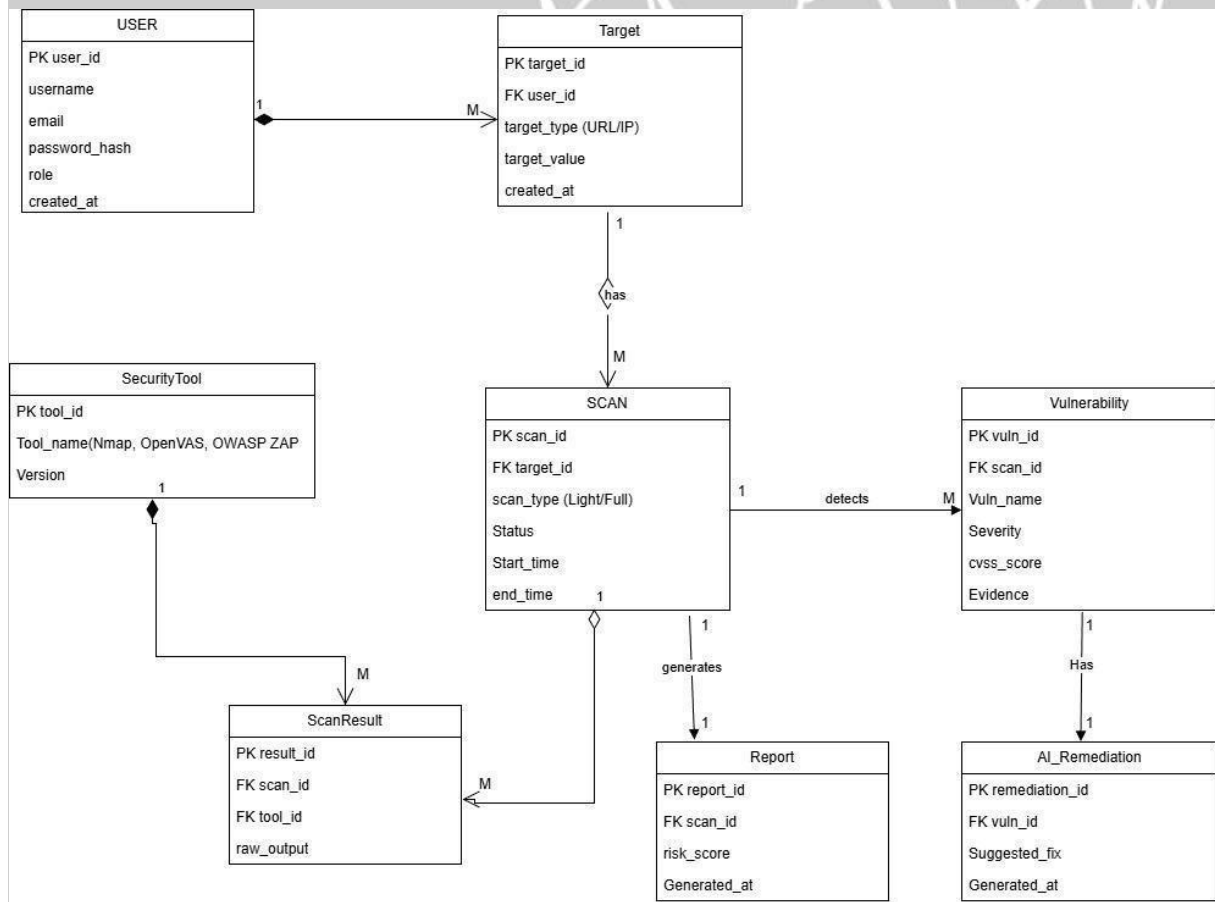


Figure 4.2: Entity Relationship Diagram (ERD)

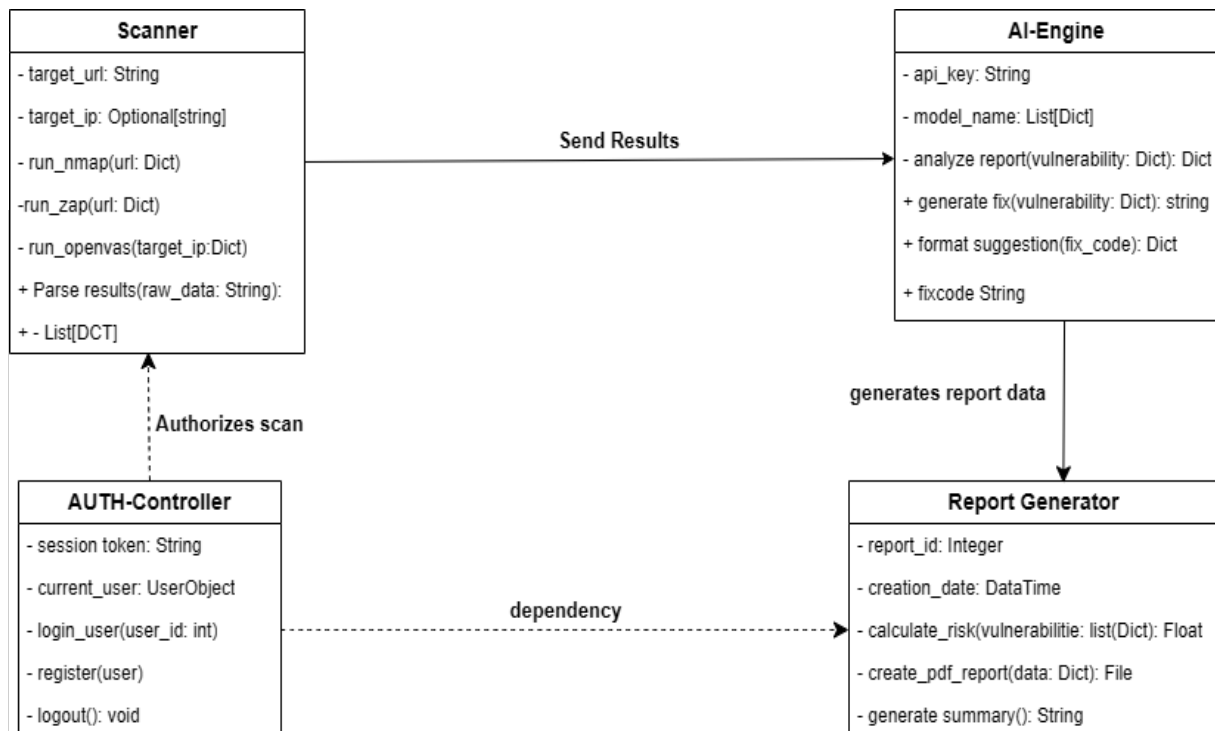


Figure 4.3: Class Diagram

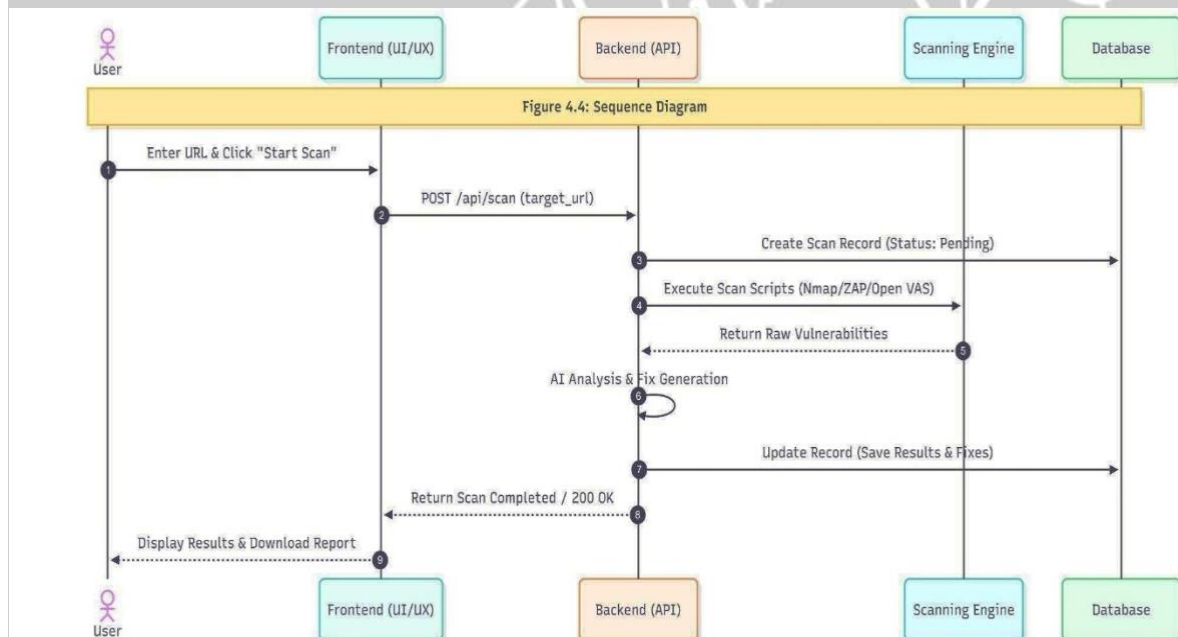


Figure 4.4: Sequence Diagram

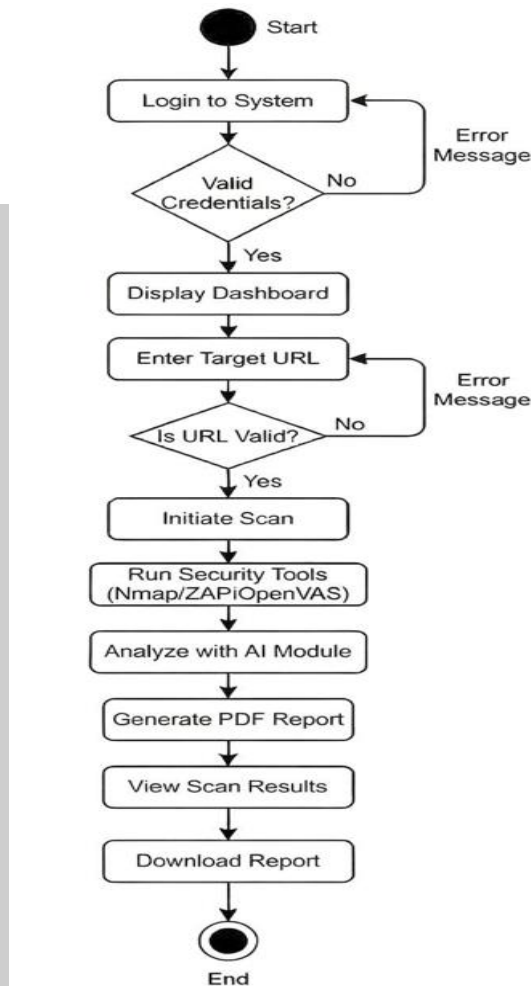


Figure 4.5: Activity Diagram

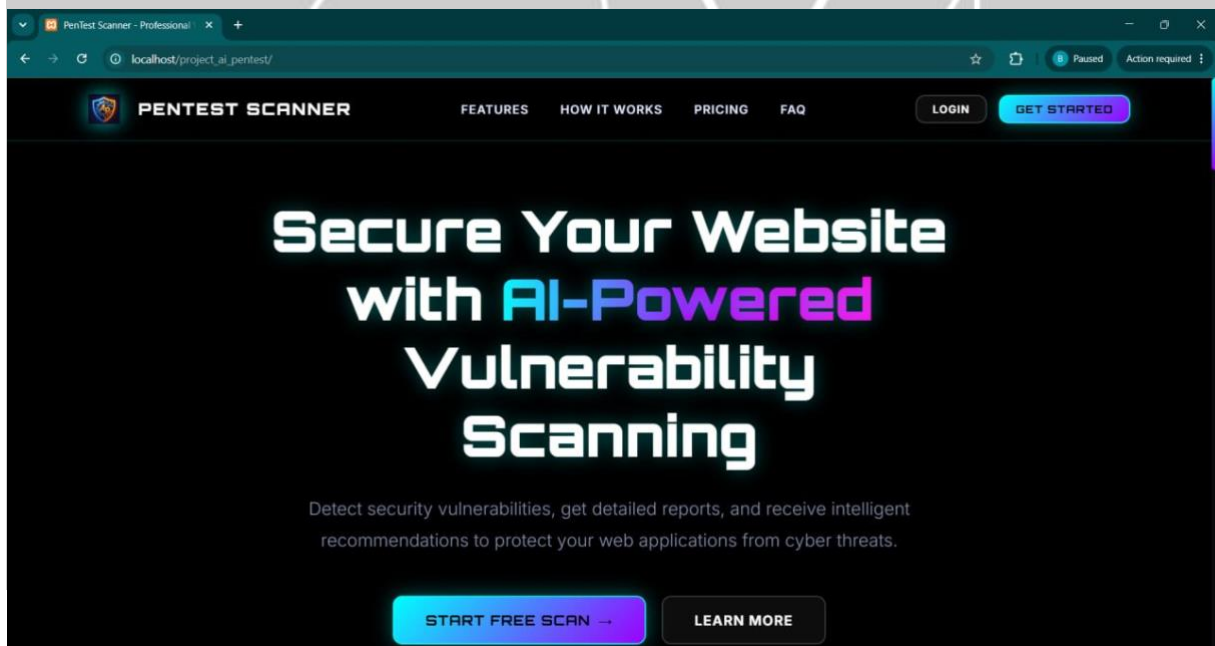


Figure 5.1: Home page

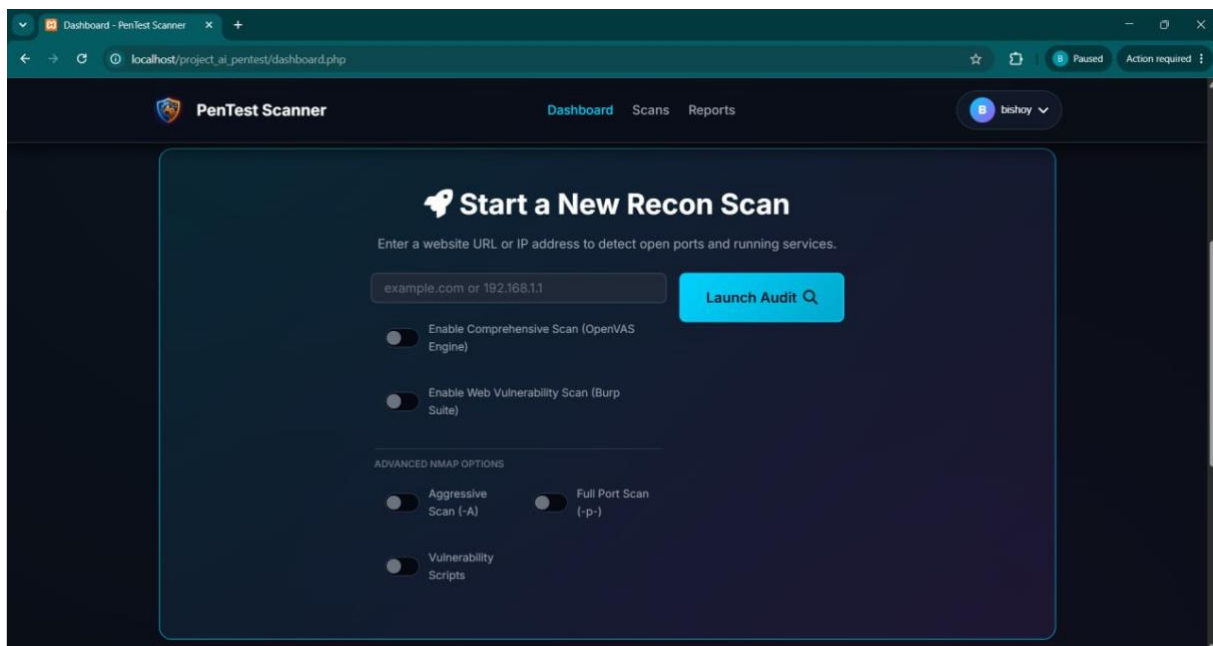


Figure 5.2: Entry page

VULNERABILITY REPORT

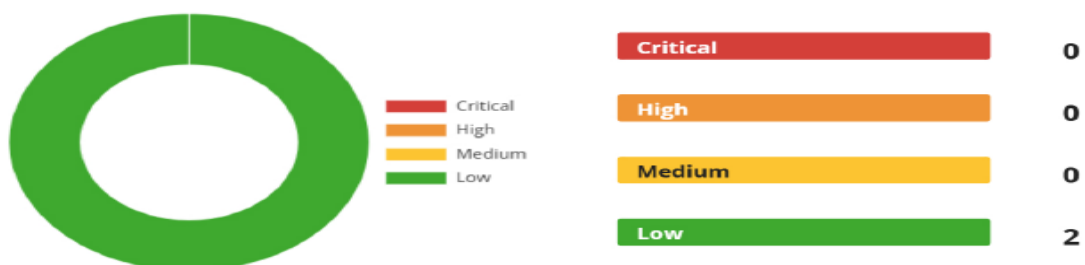
Generated by PenTest Scanner

Report Date: June 9, 2026, 10:40 pm

Scan ID: V-00090

Target Information	http://scanme.nmap.org
Scan Start Date	2026-06-09 22:39:09
Scan Duration	33 seconds
Scanner Engines	Nmap + AI Classification & RAG

Vulnerabilities by Severity

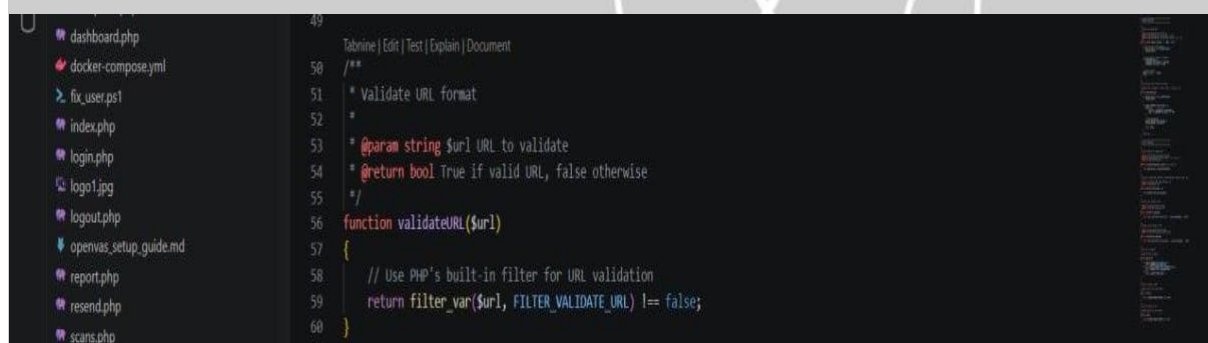


Vulnerability Details

Open Port	ID: 1
RISK FACTOR Low	
DESCRIPTION Open Port: 22/tcp (Service: ssh - OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13)	
REMEDIATION / SOLUTION Review open port 22. Ensure the service (ssh) is secure and up to date. Close the port via firewall if it is not needed publicly.	

Open Port	ID: 2
RISK FACTOR Low	
DESCRIPTION Open Port: 80/tcp (Service: http - Apache httpd 2.4.7)	
REMEDIATION / SOLUTION Review open port 80. Ensure the service (http) is secure and up to date. Close the port via firewall if it is not needed publicly.	

Figure 5.3: Vulnerability Report



```
49  
50 /**  
51  * Validate URL format  
52  *  
53  * @param string $url URL to validate  
54  * @return bool True if valid URL, false otherwise  
55  */  
56 function validateURL($url)  
57 {  
58     // Use PHP's built-in filter for URL validation  
59     return filter_var($url, FILTER_VALIDATE_URL) !== false;  
60 }
```

Figure 5.4: URL validation module



Figure 5.5: Vulnerability Scanner Module

```

login.php 34
logo1.jpg 35
logout.php 36
openvas_setup_guide.md 37
report.php 38

```

```

// 2. Get all Findings for this scan
$stmt = $pdo->prepare("SELECT * FROM scan_results WHERE scan_id = :id ORDER BY FIELD(severity, 'critical', 'high', 'medium', 'low')");
$stmt->execute(['id' => $scan_id]);
$findings = $stmt->fetchAll(PDO::FETCH_ASSOC);

```

```

openvas_wrapper.php 476
scans 477
cancel.php 478
process.php 479
status.php 480
scratch 481
update_schema.php 482
sql 483
database.sql 484
all_reports.php 485
dashboard.php 486
docker-compose.yml 487
fix_user.ps1 488
index.php 489
login.php 490
logo1.jpg 491
logout.php 492
openvas_setup_guide.md 493
report.php 494

```

```

}

// PDF Generation
document.getElementById('downloadPdfBtn').addEventListener('click', function() {
    const element = document.getElementById('reportContent');
    const chartContainer = document.getElementById('chartContainer');
    const originalText = this.innerHTML;

    this.innerHTML = '<span><i class="fas fa-spinner fa-spin"></i> Generating PDF...</span>';
    this.disabled = true;

    // TRICK: Convert Chart to Image before printing so it doesn't disappear
    let imgEl = null;
    if (chartContainer) {
        imgEl = new Image();
        imgEl.src = chartContainer.toDataURL("image/png", 1.0);
        imgEl.style.width = '100%';
        imgEl.style.maxWidth = '350px';

        chartContainer.style.display = 'none';
        chartContainer.appendChild(imgEl);
    }
});

```

Figure 5.6: Report Generation Module

```

assets 12
css 13
style.css 14
images 15
js 16
main.js 17
auth 18
authenticate.php 19
register.php 20
includes 21
PHPMailer 22
ai_wrapper.php 23
burp_wrapper.php 24
config.php 25
db.php 26
functions.php 27
nmap_wrapper.php 28
openvas_wrapper.php 29
scans 30
cancel.php 31
process.php 32
status.php 33
scratch 34
update_schema.php 35
sql 36
database.sql 37
all_reports.php 38
dashboard.php 39
docker-compose.yml 40
fix_user.ps1 41
OUTLINE 42
TIMELINE 43

```

```

// =====
// PDO DATABASE CONNECTION
// =====

try {
    // Create DSN (Data Source Name) string
    // Format: "mysql:host=localhost;dbname=database_name;charset=utf8mb4"
    $dsn = "mysql:host=" . DB_HOST . ";dbname=" . DB_NAME . ";charset=" . DB_CHARSET;

    // PDO options for better security and error handling
    $options = [
        // Set error mode to exceptions (easier to catch and handle errors)
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,

        // Return rows as associative arrays (easier to work with)
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,

        // Disable emulated prepared statements (use real prepared statements for security)
        PDO::ATTR_EMULATE_PREPARES => false,

        // Set persistent connection for better performance
        // Note: Can be set to false if you experience connection issues
        PDO::ATTR_PERSISTENT => false
    ];

    // Create PDO instance (establish connection)
    $pdo = new PDO($dsn, DB_USER, DB_PASS, $options);
    // If you reach this line, connection was successful
    // You can uncomment the line below for testing, but remove it in production
    // echo "Database connection successful!";

} catch (PDOException $e) {
    // =====
    // ERROR HANDLING
    // =====
    die("Database Connection Failed: " . $e->getMessage());
}

```

Figure 5.7: Database Connection Module

```

assets 72
css 73
style.css 74
images 75
js 76
main.js 77
auth 78
authenticate.php 79
register.php 80
includes 81
PHPMailer 82
ai_wrapper.php 83
burp_wrapper.php 84
config.php 85
db.php 86
functions.php 87
nmap_wrapper.php 88

```

```

// =====
// AUTHENTICATE USER
// =====

try {
    /**
     * Fetch user by email
     * Added 'is_verified' to check OTP activation status
     */
    $stmt = $pdo->prepare("
        SELECT id, username, email, password, is_verified, created_at
        FROM users
        WHERE email = :email
        LIMIT 1
    ");
    $stmt->execute(['email' => $email]);

    // Fetch the user record
    $user = $stmt->fetch();
}

```

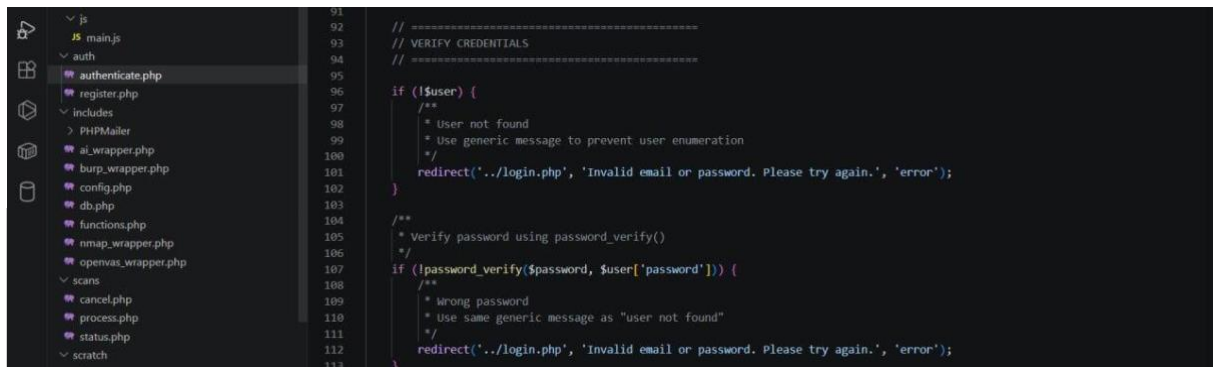



Figure 5.8: Authentication

Tables:

Table 2.1: Gantt Chart

ID	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Quarter	1st Quarter	2nd Quarter	3rd Q
1	PHASE 1: PLANNING & RESEARCH	18 days	Mon 11/3/25	Wed 11/26/25		All Team	Nov	Dec	Jan	Feb
2	Project Team Formation & Roles	1 day	Mon 11/3/25	Mon 11/3/25		Malak Abdelaziz				
3	Setup Communication	1 day	Tue 11/4/25	Tue 11/4/25	2	Bishoy Gerges				
4	Define Project Scope & Objectives	2 days	Wed 11/5/25	Thu 11/6/25	3	Malak, Bishoy				
5	Identify Key Stakeholders	1 day	Fri 11/7/25	Fri 11/7/25	4	Malak Abdelaziz				
6	Conduct Feasibility Study	2 days	Fri 11/7/25	Mon 11/10/25	5	Abdelrahman, Mahmoud				
7	Research Web Scanners (ZAP)	6 days	Tue 11/11/25	Tue 11/18/25	6	Bishoy, Mohamed Hussein				
8	Research Network Scanners (Nmap)	6 days	Tue 11/11/25	Tue 11/18/25	6	Mahmoud, Rania Omar				
9	Research AI Models for Fixes	6 days	Tue 11/11/25	Tue 11/18/25	6	Mohamed Ashraf, Nada				
10	Compare & Select Tech Stack	2 days	Wed 11/19/25	Thu 11/20/25	7, 8, 9	Bishoy, Abdelrahman				
11	Define Hardware/Software Req.	1 day	Fri 11/21/25	Fri 11/21/25	10	Nada, Rania				
12	Draft Project Proposal	2 days	Mon 11/24/25	Tue 11/25/25	11	Malak, Mohamed Hussein				
13	Review & Submit Proposal	1 day	Wed 11/26/25	Wed 11/26/25	12	Malak Abdelaziz				
14	PHASE 2: ANALYSIS (TERM 1)	23 days	Thu 11/27/25	Mon 12/29/25		Analysts				
15	Gather Functional Requirements	4 days	Thu 11/27/25	Tue 12/2/25	13	Malak, Rania Omar				
16	Analyze Non-Functional	2 days	Wed 12/3/25	Thu 12/4/25	15	Abdelrahman, Mahmoud				
17	Perform Risk Analysis	2 days	Fri 12/5/25	Mon 12/8/25	16	Malak, Bishoy				
18	Define User Roles & Permissions	1 day	Wed 12/3/25	Wed 12/3/25	17	Bishoy Gerges				
19	Define AI Model Data Sources	3 days	Wed 12/3/25	Fri 12/5/25	18	Mohamed Ashraf, Nada				
20	Create Context Diagram (DFD 10)	2 days	Thu 12/4/25	Fri 12/5/25	19	Mohamed Hussein				
21	Create DFD Level 1 (Detailed)	3 days	Mon 12/8/25	Wed 12/10/25	20	Mohamed Hussein, Nada				
22	Create Use Case Diagrams	2 days	Thu 12/4/25	Fri 12/5/25	18	Rania, Mohamed Ashraf				
23	Create Functional Requirements Doc (SRS)	4 days	Mon 12/8/25	Thu 12/11/25	22	Mohamed Hussein				
24	Write Use Case Scenarios	3 days	Mon 12/8/25	Wed 12/10/25	22	Rania, Malak				
25	PHASE 3: DESIGN (CRITICAL FOR)	13 days	Thu 12/11/25	Mon 12/29/25		Designers				
26	Design Conceptual Data	2 days	Thu 12/11/25	Fri 12/12/25	24	Mahmoud, Bishoy				
27	Design ERD (Database)	3 days	Mon 12/15/25	Wed 12/17/25	26	Mahmoud, Abdelrahman				
28	Design System Architecture	2 days	Thu 12/18/25	Fri 12/19/25	27	Bishoy Gerges				
29	Design Network Topology (Nmap)	3 days	Mon 12/22/25	Wed 12/24/25	28	Mahmoud Hamdy				
30	Design Security Architecture	3 days	Thu 12/25/25	Mon 12/29/25	29	Bishoy, Abdelrahman				
31	Design Class Diagrams	3 days	Thu 12/18/25	Mon 12/22/25	27	Abdelrahman Gamal				
32	Design Sequence Diagrams	3 days	Thu 12/18/25	Mon 12/22/25	27	Nada Mohamed				
33	Design UI Wireframes (Login/Reg)	3 days	Thu 12/11/25	Mon 12/15/25	24	Rania, Mohamed Hussein				
34	Design Dashboard	3 days	Tue 12/16/25	Thu 12/18/25	33	Bishoy, Rania Omar				
35	Design Report Interface	2 days	Fri 12/19/25	Mon 12/22/25	34	Mohamed Hussein				
36	Design UI Style Guide	2 days	Tue 12/23/25	Wed 12/24/25	35	Rania Omar				
37	PHASE 4: DOCS WRITING (CH 1, 2, 3,)	34 days	Thu 11/27/25	Tue 1/13/26		All (Split)				
38	Write Chapter 1 (Introduction)	5 days	Thu 11/27/25	Wed 12/3/25	37	Malak, Nada Mohamed				
39	Write Chapter 2 (Planning)	5 days	Thu 11/27/25	Wed 12/3/25	37	Abdelrahman, Mahmoud				

ID	Task Mo	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Timeline												
								Quarter	1st Quarter	2nd Quarter	3rd Q									
								Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul				
40		Write Chapter 3 (Analysis)	5 days	Tue 12/30/25	Mon 1/5/26	24,24	Malak,Mohamed Ashraf			Malak,Mohamed Ashraf										
41		Write Chapter 4 (System Design)	7 days	Tue 12/30/25	Wed 1/7/26	30,32,30,29,29	Nada,Rania			Nada,Rania										
42		Review All Chapters (1, 2, 3, 4)	4 days	Thu 1/8/26	Tue 1/13/26	38,39,40,41	Malak,Bishoy			Malak,Bishoy										
43		SUBMIT DOCUMENTATION (CH 1-4)	0 days	Tue 1/13/26	Tue 1/13/26	42	All Team			1/13										
44		PHASE 5: FRONTEND PROTOTYPE	27 days	Fri 12/12/25	Mon 1/19/26		Frontend Team													
45		Setup Git Repository for	1 day	Fri 12/12/25	Fri 12/12/25	10,19,23	Bishoy Gerges			Bishoy Gerges										
46		Create Project Structure	1 day	Mon 12/15/25	Mon 12/15/25	49	Mohamed Hussein			Mohamed Hussein										
47		Develop Global Navigation Bar	2 days	Tue 12/16/25	Wed 12/17/25	48	Rania Omar			Rania Omar										
48		Develop Footer Component	1 day	Thu 12/18/25	Thu 12/18/25	47	Mohamed Hussein			Mohamed Hussein										
49		Develop Registration Page	3 days	Thu 12/18/25	Mon 12/22/25	47	Mohamed Hussein			Mohamed Hussein										
50		Develop Login Page	3 days	Thu 12/18/25	Mon 12/22/25	47	Rania Omar			Rania Omar										
51		Develop Dashboard Sidebar Menu	2 days	Tue 12/23/25	Wed 12/24/25	50	Bishoy Gerges			Bishoy Gerges										
52		Develop "New Scan" Input Form	3 days	Thu 12/25/25	Mon 12/29/25	51	Mohamed Hussein			Mohamed Hussein										
53		Develop "Scan History" Data Table	3 days	Thu 12/25/25	Mon 12/29/25	51	Rania Omar			Rania Omar										
54		Implement Static Charts (Chart.js)	3 days	Tue 12/30/25	Thu 1/1/26	53	Rania,Bishoy			Rania,Bishoy										
55		Refine UI: Make Responsive	5 days	Fri 1/2/26	Thu 1/8/26	54	Rania,Mohamed Hussein			Rania,Mohamed Hussein										
56		Deliver UI Prototype	0 days	Thu 1/8/26	Thu 1/8/26	59	Frontend Team			1/8										
57		Mid-Year Exams Break (Limited Work on Docs)	18 days	Thu 12/25/25	Mon 1/19/26		-			-										
58		PHASE 6: BACKEND DEV (TERM 2)	34 days	Mon 1/19/26	Thu 3/5/26		Backend Team													
59		Setup Python Flask Virtual Env	1 day	Tue 1/20/26	Tue 1/20/26	61	Mahmoud Hamdy			Mahmoud Hamdy										
60		Install Project Dependencies	1 day	Wed 1/21/26	Wed 1/21/26	59	Abdelrahman Gamal			Abdelrahman Gamal										
61		Configure Git for Backend	1 day	Mon 1/19/26	Mon 1/19/26	43	Bishoy Gerges			Bishoy Gerges										
62		Initialize MySQL Database	2 days	Thu 1/22/26	Fri 1/23/26	60	Abdelrahman Gamal			Abdelrahman Gamal										
63		Module 1: Authentication & Authorization APIs	4 days	Mon 1/26/26	Thu 1/29/26	62	Mahmoud Hamdy			Mahmoud Hamdy										
64		Develop User Profile APIs	3 days	Fri 1/30/26	Tue 2/3/26	63	Abdelrahman Gamal			Abdelrahman Gamal										
65		Module 2: Scanner Integration (Nmap/ZAP)	8 days	Mon 1/26/26	Wed 2/4/26	62	Bishoy,Mahmoud			Bishoy,Mahmoud										
66		Script: OWASP ZAP API Client	8 days	Mon 1/26/26	Wed 2/4/26	62	Abdelrahman,Nada			Abdelrahman,Nada										
67		Script: OpenVAS Scanner Integration	8 days	Mon 1/26/26	Wed 2/4/26	62	Nada,Mohamed Hussein			Nada,Mohamed Hussein										
68		Implement Nmap Result Parser	4 days	Thu 2/5/26	Tue 2/10/26	69	Mahmoud Hamdy			Mahmoud Hamdy										
69		Implement ZAP Result Parser	4 days	Thu 2/5/26	Tue 2/10/26	68	Abdelrahman Gamal			Abdelrahman Gamal										
70		Develop Vulnerability Risk	4 days	Wed 2/11/26	Mon 2/16/26	68,69	Bishoy Gerges			Bishoy Gerges										
71		Create PDF Report Generation Logic	5 days	Tue 2/17/26	Mon 2/23/26	70	Abdelrahman Gamal			Abdelrahman Gamal										
72		Module 3: Logging & Reporting System	5 days	Tue 2/17/26	Mon 2/23/26	70	Abdelrahman,Nada			Abdelrahman,Nada										
73		Create Scan Request/Result	4 days	Tue 2/24/26	Fri 2/27/26	71	Mahmoud,Nada			Mahmoud,Nada										
74		Backend Error Handling & Logging	3 days	Mon 3/2/26	Wed 3/4/26	73	Bishoy,Mahmoud			Bishoy,Mahmoud										
75		Backend Code Review & Cleanup	1 day	Thu 3/5/26	Thu 3/5/26	74	Bishoy Gerges			Bishoy Gerges										
76		System Ready for Integration	0 days	Thu 3/5/26	Thu 3/5/26	79	Abdelrahman Gamal			3/5										

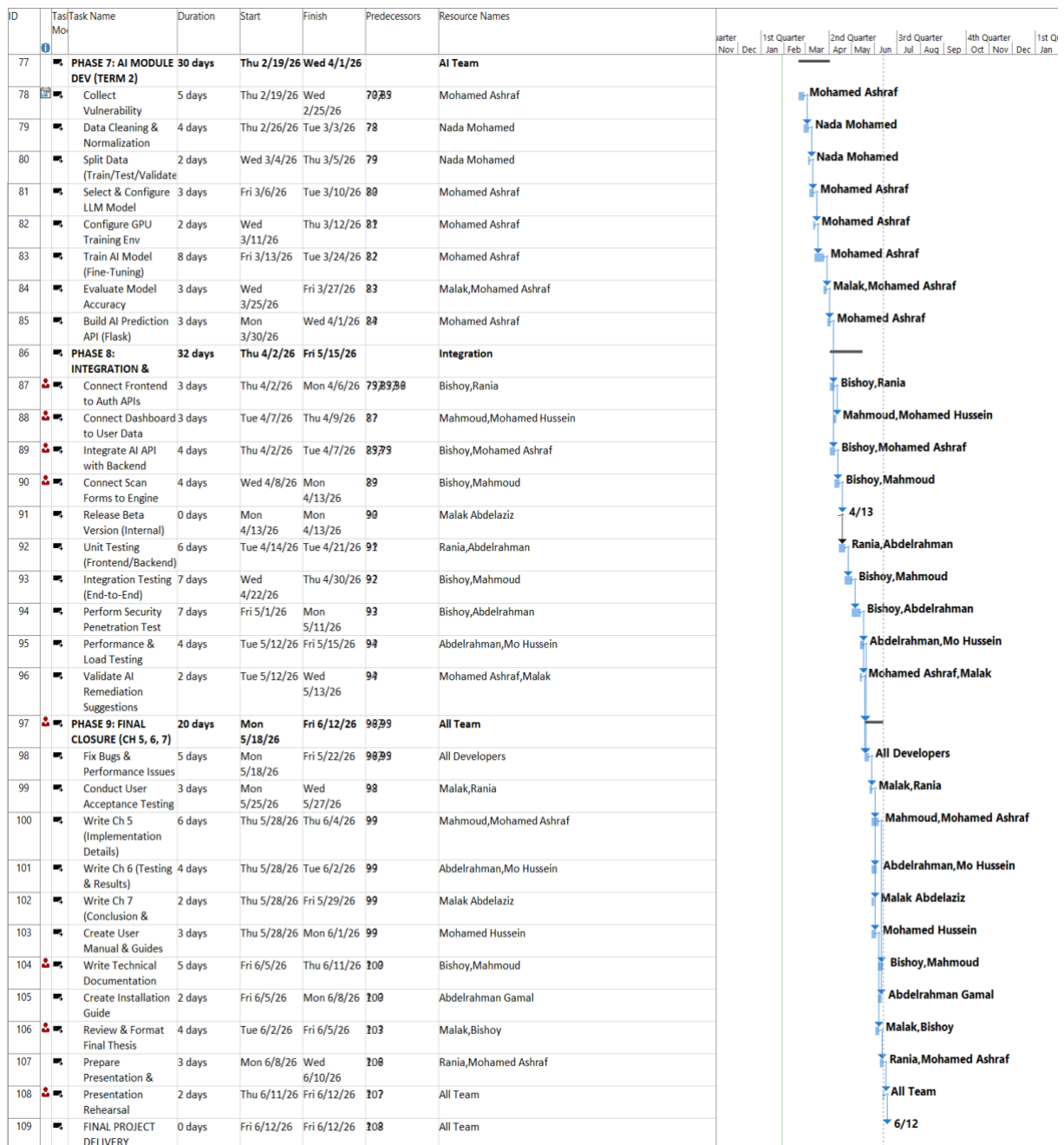


Table 2.1: SWOT Analysis Summary

Category	Description
Strengths	Automation, AI integration, and cost-effectiveness.
Weaknesses	False positives and system complexity.
Opportunities	Cloud security expansion and DevSecOps integration.
Threats	Evolving threats and legal risks.

Table 6.1: System Testing Results

Test Case	Expected Result	Actual Result	Status
User Registration	User account is created successfully	Account created successfully	Pass
User Login	User logs in successfully	Login completed successfully	Pass
Authentication Code Sending	Authentication code is sent to the user's email	Code sent successfully	Pass
URL Validation	Accept valid URLs and reject invalid URLs	Validation performed correctly	Pass
Vulnerability Scanning	Scan starts and detects vulnerabilities	Scan executed successfully	Pass
Vulnerability Display	Detected vulnerabilities are displayed correctly	Vulnerabilities displayed successfully	Pass
Report Generation	Security report is generated correctly	Report generated successfully	Pass
AI Analysis	AI provides recommendations based on scan results	Recommendations generated successfully	Pass
Database Connection	Data is stored and retrieved correctly	Database operations successful	Pass
Integration with Nmap, Burp Suite, and OpenVAS	Scanning tools communicate successfully with the system	Integration completed successfully	Pass
Account Activation	User account is activated after entering the authentication code	Account activated successfully	Pass

Table 7.1: Future Work Plan

No.	Proposed Enhancement	Description
1	Support Additional Scanning Tools	Integrate more vulnerability scanning tools and security frameworks to improve detection capabilities.
2	Enhanced AI Analysis	Improve the AI module to provide more accurate vulnerability analysis and remediation recommendations.
3	Real-Time Monitoring	Implement continuous security monitoring and real-time vulnerability detection.
4	Security Dashboard	Develop a dashboard to track previous scans, vulnerabilities, and security statistics.
5	Automated Remediation	Add features that can automatically fix or mitigate certain types of vulnerabilities.
6	Advanced Reporting	Enhance report generation with charts, risk metrics, and detailed security insights.
7	Mobile Application Testing	Extend the system to support penetration testing for mobile applications.
8	Cloud Deployment	Deploy the system on cloud platforms to improve scalability, availability, and performance.

2.6. Future Work

No.	Proposed Enhancement	Description
1	Support Additional Scanning Tools	Integrate more vulnerability scanning tools and security frameworks to improve detection capabilities.
2	Enhanced AI Analysis	Improve the AI module to provide more accurate vulnerability analysis and remediation recommendations.
3	Real-Time Monitoring	Implement continuous security monitoring and real-time vulnerability detection.
4	Security Dashboard	Develop a dashboard to track previous scans, vulnerabilities, and security statistics.
5	Automated Remediation	Add features that can automatically fix or mitigate certain types of vulnerabilities.
6	Advanced Reporting	Enhance report generation with charts, risk metrics, and detailed security insights.
7	Mobile Application Testing	Extend the system to support penetration testing for mobile applications.
8	Cloud Deployment	Deploy the system on cloud platforms to improve scalability, availability, and performance.

2.7. Project Members Contribution

Student Name	Attendance (20%)	Technical Contribution (30%)	Task Completion (20%)	Teamwork (15%)	Communication (15%)	Total (%)
Member 1						
Member 2						
Member 3						
Member 4						
Member 5						
Member 6						
Member 7						
Member 8						

Criterion	Description	Weight
Attendance & Commitment	Attending meetings and project activities	20%
Technical Contribution	Coding, design, testing, documentation, research	30%
Task Completion	Completing assigned tasks on time	20%
Teamwork	Cooperation and support of team members	15%
Communication	Reporting progress and participating in discussions	15%

Assessment Criteria